Dhruv Ramani

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Summary

Third-year B.Tech Computer Science & AI student who ships end-to-end ML systems rather than notebooks. Built a physics-informed exoplanet detection pipeline over live NASA TESS data (calibrated RF + XGBoost ensemble, 0.957 held-out ROC-AUC) and wrote up a cross-mission characterization study of it; shipped a signed Android build of an 11-language, voice-first health assistant serving a 211k-facility offline database. Comfortable across applied ML, computer vision, and full-stack delivery.

Education

Newton School of Technology (Rishihood University) India
B.Tech in Computer Science and Artificial Intelligence (3rd Year) *Aug 2024 – 2028*

- **Relevant Coursework:** Neural Networks and Deep Learning, Generative AI, Linear Algebra, Statistics, Data Analytics and Visualization

Projects

Exoplanet Detection Pipeline | *Python, Scikit-learn, XGBoost, FastAPI, Next.js, Streamlit* Feb 2025 – Present

- Built an end-to-end transit-detection pipeline over live NASA TESS data: SPOC lightcurves stitched from MAST (with a TESScut FFI photometry fallback), `wotan` biweight detrending, and a Transit Least Squares period search seeded with TIC stellar priors.
- Trained a soft-voting RandomForest + XGBoost ensemble on 7,325 labeled Kepler objects of interest with Platt calibration on a held-out split: **89.8% accuracy, 0.957 ROC-AUC, 0.917 PR-AUC, 0.080 Brier** on 1,465 unseen targets.
- Layered five physics vetting checks over the classifier – odd/even depth consistency (Welch's t-test), transit-implied stellar density, duration vs. circular-orbit maximum, secondary-eclipse search, and an SDE floor – and emitted a plain-English reasoning trail for every verdict.
- Authored a characterization study (preprint in repo) quantifying **Kepler→TESS domain shift** on a 500-TOI benchmark: ROC-AUC degrades 0.96 → 0.72 [0.67, 0.76]; recall recovers 0.28 → 0.72 via a cross-validated decision threshold, validated with injection–recovery completeness maps and a label-permutation null test.

Swasthya AI – Multilingual Health Assistant | *Flutter, Groq/Llama 3.3, Gemini Vision, SQLite, Supabase* Jan 2026 – Present

- Shipped a signed Android build (∼21.7k lines of Dart, zero `flutter analyze` issues) of a voice-first health assistant covering **11 Indian languages**, with token-by-token SSE streaming from a pluggable LLM backend.
- Implemented emergency red-flag detection using inflection-safe stem matching across all 11 languages plus romanized Hindi, which suppresses the model answer entirely and surfaces a one-tap 112 dialer intent.
- Bundled **211,200 government health facilities** as a compressed SQLite database for fully offline, distance-sorted lookup, and added Gemini-vision lab-report and medicine-strip scanning with an OCR-only prompt contract.
- Hardened the release path: all AI and feedback traffic proxied through a Supabase Edge Function so no credential ships in the APK; health profile and symptoms held in Android Keystore / iOS Keychain; a circuit breaker (2 failures → 10-minute skip) fronts a custom fine-tuned Qwen2.5-7B LoRA backend.

Mavis – Real-Time CV Fitness Coach | *Python, MediaPipe, OpenCV, React 19, TypeScript, pytest* Nov 2025 – Present

- Engineered an on-device pose pipeline – 33 MediaPipe landmarks → EMA smoothing → a 3-second stillness-gated calibration → geometric features → a per-exercise finite state machine that counts reps and scores each one out of 100. No video ever leaves the device.
- Derived body-relative fault detectors (elbow drift normalized by the user's own shoulder width, bilateral asymmetry, tempo floor, torso swing) so that shifting stance mid-set is not misread as a form fault.
- Cut the false-fault rate from **61%→5%** (shoulder press) and **46%→10%** (bicep curl) by setting every tolerance from measured distributions over 128 recorded reps, adding graded penalties and multi-frame fault debouncing.
- Unified the Python and TypeScript runtimes behind a single `exercises.json` threshold spec, with a pytest suite that replays recorded landmark fixtures on a virtual clock and asserts both that clean reps pass and that poor form is still caught.

Technical Skills

Languages: Python, JavaScript/TypeScript, Dart, SQL
ML / DL: PyTorch, Scikit-learn, XGBoost, Hugging Face Transformers, LoRA/PEFT fine-tuning, model calibration
Web & Mobile: React.js, Next.js, FastAPI, Node.js, Flutter, Streamlit, Tailwind CSS
Libraries & Tools: NumPy, Pandas, OpenCV, MediaPipe, SQLite, Supabase, Git, pytest, LaTeX